Who Stole My Genes?

Detecting HGT Candidates Using Graph-Based Analysis

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1 Introduction

1.1 Background

Horizontal Gene Transfer (HGT) is a major evolutionary mechanism in bacteria that enables genes to move between organisms independently of vertical inheritance. This process allows bacteria to share functional genes directly with other living organisms, rather than relying solely on reproduction and inheritance to pass genetic traits to the next generation.[1] This can create cases where a gene in one species is unusually similar to a gene in a phylogenetically distant species. However, high similarity across distant taxa is not unique to HGT; such patterns may

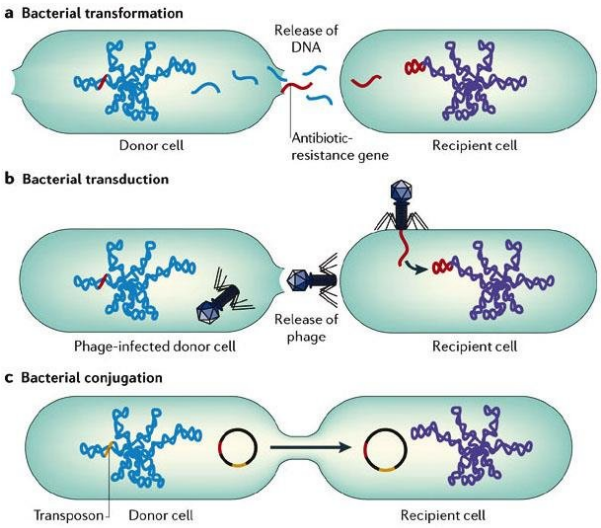

Figure 1: Biological mechanisms underlying horizontal gene transfer (HGT).
From *Nature Microbiology, © 2006 Nature Publishing Group*

also result from strong functional conservation, heterogeneous evolutionary rates, gene loss in intermediate lineages, or annotation and assembly artifacts.[2] Previous studies have identified numerous cases of HGT involving protein-coding genes in bacteria, most notably antibiotic resistance genes. In addition, transfer of virulence factors, toxins, and genes involved in stress response and metabolism is also prevalent.[3] All these give the recipient organism a critical advantage in stressful environments.

Detecting HGT remains computationally challenging. Established approaches either rely on phylogenetic incongruence (comparing gene trees to species trees), which is computationally heavy and sensitive to alignment quality, or use simpler heuristics that can be noisy.[4]
Our project explores a complementary, lightweight approach: represent cross-species gene similarities as a graph and use global graph patterns along with species-distance context to prioritize candidate HGT genes.

1.2 Research Question

How can graph-based analysis of gene similarity reveal candidate horizontal gene transfer events across bacterial genomes?

1.3 Hypothesis

We hypothesize that genes involved in horizontal gene transfer will exhibit disproportionately high similarity to phylogenetically distant species, forming graph edges that deviate from expected species-level clustering.

1.4 Objectives

- Construct a sparse gene similarity graph using protein sequences
- Partition proteins into homologous groups and construct within-group similarity graphs
- Define and compute an anomaly score that highlights unusually high similarity across large phylogenetic distances
- Generate interpretable visualizations
- Validate candidates by cross-referencing known HGT-related genes in the literature

2 Data

Our project was split into 2 methodologies. Both methodologies used data from NCBI RefSeq,[5] and the NCBI Taxonomy database[6] (using `BioPython`'s `Entrez` module[7]). For each genome, we acquired FASTA files (`.faa`) where each entry corresponds to a single gene.
For the first methodology, we downloaded the `bacteria.*.protein.faa.gz` files from the bacteria database,[8] which consist of 15 bacterial proteomes.
For the second methodology, we acquired protein sequences of 48 species spanning 19 taxonomic families from the genome database,[9] using a python script (`refseq_fetch_proteins.py`). To control dataset size, we selected up to two assemblies per species using a deterministic ranking. While this choice ensured manageable scale and consistent annotation quality, it introduced some redundancy due to near-identical assemblies (we discuss this limitation below).

3 Methods

Our work was split into 2 methodologies, both aimed to find HGT candidates using graph-based analysis, but from different perspectives. The first method focused at the protein level, and the second at the proteome level. Instead of MSA and computationally intensive similarity calculations on entire proteomes, both methods employ a divide-and-conquer approach to ease computation.

3.1 Method 1 – Identifying HGT Candidates Using Orthologous Clustering

This method seeks to locate similarly functional genes (orthologs) using sequence similarity clustering, and analyze each orthogonal group separately. The underlying assumption is that genes involved in HGT will cluster with functionally similar sequences across species, reducing the need for excessive similarity computations. This method consists of four main steps, as described in fig. 2.

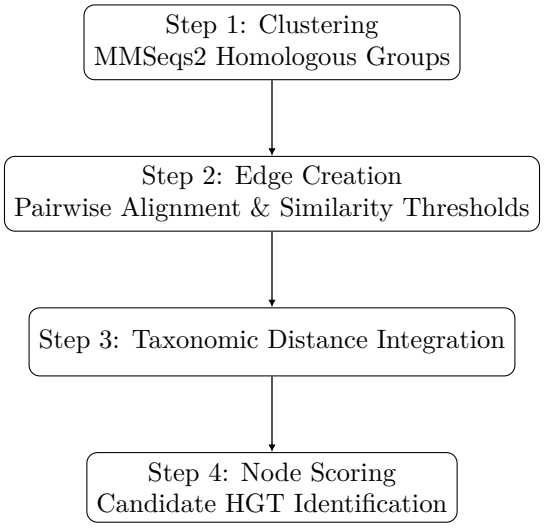

Figure 2: Workflow of Method 1. Proteins are first clustered into homologous groups, followed by within-cluster alignment to construct similarity edges. Taxonomic distance is incorporated to detect anomalous cross-species similarities, and node-level scoring is used to identify candidate HGT genes.

3.1.1 Clustering

As discussed in the data section, we downloaded protein sequences of 15 bactrial proteomes. We then clustered all proteins into homologous groups using MMseqs2.[10] We used a minimum sequence identity threshold of 50% and a coverage threshold of 80% (MMseqs2 parameters: `--min-seq-id 0.5 -c 0.8`), so that sequences grouped together are broadly similar and likely to share related function.

3.1.2 Edge Creation Based on Similarity Score

Within each homologous cluster, we computed pairwise sequence alignments between all unordered protein pairs. Alignments were performed using the `PairwiseAligner` class from `Biopython` (v1.85), configured for local alignment (mode `local`) with the BLOSUM62 substitution matrix.[11] Gap penalties were set to a gap opening score of -10 and a gap extension score of -1. For each aligned pair (u, v), we computed the following metrics:

$$\text{Identity}(u, v) = \frac{\#\text{matching aligned residues}}{\min\{\text{len}(u), \text{len}(v)\}}$$

$$\text{Coverage}(u, v) = \frac{\text{aligned_length}}{\min\{\text{len}(u), \text{len}(v)\}}$$

An undirected edge between u and v was created only if:

$$\text{Identity} \geq 0.5, \quad \text{Coverage} \geq 0.8, \quad \text{aligned_length} \geq 50$$

The minimum aligned length of 50 residues was chosen to avoid spurious edges arising from short conserved motifs or domain fragments. For each edge, we defined the similarity score as:

$$\text{SimilarityScore}(u, v) = \text{Identity}(u, v) \times \text{Coverage}(u, v).$$

For a cluster of n proteins, this requires $O(n^2)$ pairwise alignments; since $n \ll N_{\text{total}}$ (where

N_{total} is the total number of proteins) after clustering, this is feasible. For pseudo-code, see algorithm 1.

Algorithm 1 BuildEdgesFromAlignments

```
Input: Dictionary seqs
Output: List edges
ids ← sorted keys of seqs
edges ← ∅
for all unordered pairs (u, v) in ids do
    (identity, coverage, aligned_len) ← align(seqs[u], seqs[v])
    if identity ≥ 0.5 and coverage ≥ 0.8 and aligned_len ≥ 50 then
        score ← identity × coverage
        Add edge (u, v) with attributes (score, identity, coverage, aligned_len) to edges
    end if
end for
return edges
```

3.1.3 Incorporating Taxonomic Information and Computing Distance

To contextualize cross-species similarities, we retrieved taxonomic lineage information for each species represented in the graph using the NCBI Taxonomy database.[6] Each graph node was annotated with taxonomic attributes corresponding to the ordered rank list:

(species, genus, family, order, class, phylum, superkingdom)

For two species u and v, we define their taxonomic distance based on the most specific (lowest) rank at which they share the same annotation. Formally, let the ordered rank list be indexed from 0 (species) to $\mathcal{R} - 1$ (superkingdom). The taxonomic distance $d(u, v)$ is defined as:

$$d(u, v) = \min \{i \mid u \text{ and } v \text{ share the same label at rank } i\}$$

If the two species do not share any common annotation across the considered ranks, we set $d(u, v) = \mathcal{R}$. Thus, smaller values of $d(u, v)$ indicate closer phylogenetic relatedness, while larger values correspond to more distant taxa.

3.1.4 Node Scoring and Identification of HGT Candidates

We flagged an edge (u, v) as suspicious if it simultaneously connected distant taxa and highly similar proteins:

$$d_{\text{tax}}(u, v) \geq 2 \quad \textbf{and} \quad \text{SimilarityScore}(u, v) \geq 0.6$$

where $d_{\text{tax}}(u, v)$ is the taxonomic distance from step 3, and SimilarityScore is defined in step 2. Intuitively, such edges represent unexpected cross-species similarity (large taxonomic distance, yet high sequence similarity).

Let $S(i)$ denote the set of nodes connected to i by suspicious edges. We then compute two quantities for every node i:

1. **HGT score** (sum of similarity scores over suspicious edges):

$$\text{score}(i) = \sum_{j \in S(i)} \text{SimilarityScore}(i, j)$$

2. **Suspicious degree** (number of suspicious edges incident to i):

$$\deg_{\text{sus}}(i) = |S(i)|$$

To select candidate HGT proteins, we considered only nodes with $\deg_{\text{sus}}(i) \geq 2$ and whose score exceeded a threshold that we needed to determine. We found the 70th percentile of the non-zero node scores (denoted as τ_{score}) and retained nodes satisfying

$$\text{score}(i) \geq \tau_{\text{score}} \quad \textbf{and} \quad \deg_{\text{sus}}(i) \geq 2$$

We ranked candidates by score and reported the top-ranked node(s). In our experiments we output the top 1 candidate(`TOP_HGT_N`= 1). i.e. the node with the highest HGT score, which also could (but not necessarily) be the gene donor.
Finally, we supported the top candidate using phylogenetic reconstruction. Phylogenetic trees were constructed with the Neighbor-Joining (NJ) algorithm[12] using the pairwise distance $d(u, v) = 1 - \text{Identity}(u, v)$, where $\text{Identity}(u, v)$ is computed in step 2. If a pairwise identity was unavailable, we set $d(u, v) = 1$.

3.2 Method 2 – Alignment-Free Detection of HGT Candidates

This method seeks to identify candidate HGT genes by constructing a similarity graph over entire proteomes using an alignment-free approach based on k-mer representations. An overview of the method is shown in fig. 3.

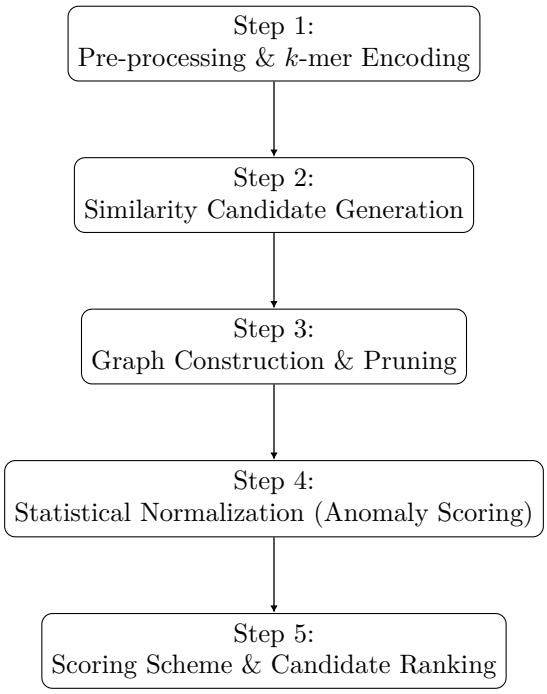

Figure 3: Workflow of Method 2. Protein sequences are represented using k-mers to generate cross-species similarity candidates. Pre-pruning controls graph density by retaining high-confidence edges, after which normalization and anomaly scoring are applied to identify candidate HGT-related proteins.

3.2.1 Pre-processing and Similarity Candidate Generation

To avoid computationally infeasible all-vs-all alignments, proteins shorter than 50 amino acids were excluded. The remaining sequences were represented as sets of distinct k-mers ($k = 6$), encoded as collision-free base-20 integers over the canonical amino acid alphabet.[13,14] The purpose of this encoding is to map each distinct k-mer to the proteins in which it appeared.
To suppress non-informative overlaps, highly frequent k-mers (appearing in more than 100 proteins) were excluded. Only cross-species pairs were considered, as intra-species similarity is expected and not informative for HGT detection, and candidate lists were capped to the top 10 matches per protein. Exact Jaccard similarity over the k-mer sets was then calculated for these candidate pairs to obtain similarity scores.

3.2.2 Graph Construction and Pruning

Candidate pairs of protein sequences were assembled into an undirected weighted similarity graph, where each node represents a protein sequence. Edges between proteins are weighted by the previously computed Jaccard similarity. Retained edges were required to meet a minimum Jaccard similarity value of 0.1, and per-node degree was capped at 20. Additional taxonomic distance filters retained edges with taxonomic distance in the range $[2, 4]$, that was calculated as shown in Method 1's step 3. These steps reduced the graph from approximately 9×10^5 candidate edges to 92,790 edges, yielding the sparse graph used to detect HGT events.

3.2.3 Graph-Based Anomaly Scoring

For each unordered species pair, we computed the median and median absolute deviation (MAD) of the edge Jaccard similarity distribution to define robust z-scores, using only pairs with at least 30 edges.
Within each connected component of the pruned graph, taxonomic mixing was quantified using Shannon entropy[15] of species labels, and anomaly concentration was defined as the fraction of edges with $z \geq 3$.
Each node was annotated with anomaly evidence (max and top-k mean z-scores) and structural connectivity metrics: cross-species degree, neighbor entropy, participation coefficient,[16] clustering coefficient,[17] and betweenness centrality.
Proteins were ranked using a component-aware scoring framework that combined several graph-derived indicators. For each protein, we considered: (i) the size of its connected component, (ii) taxonomic mixing within that component measured as the Shannon entropy of species labels, (iii) the node's cross-species degree in the similarity graph, and (iv) the mean of the five largest z-scores among its incident edges. Candidates required minimum thresholds (component size $\geq$ 10, entropy ≥ 0.75, node degree ≥ 4, mean top-5 z ≥ 2).
To rank candidate proteins, we define a final anomaly score $S(u)$ for each protein u, combining component-level and edge-level signals. The score is computed as:

$$\log S(u) = \log \Phi(C_u) + \log A(u) + \log T(u)$$

Here $\Phi(C_u)$ captures component-level taxonomic mixing and anomaly concentration, $A(u)$ aggregates normalized similarity scores, and $T(u)$ summarizes cross-species connectivity. Aggregation in log-space ensures high scores arise only when anomaly and structural evidence co-occur.

4 Results

4.1 Constructed Graph

In method 2, we created a candidate graph which can be used to find HGT events.
The full candidate graph contains 911,306 cross-species edges over 214,136 proteins. After Jaccard-based pre-pruning it contains 92,790 edges over 37,421 proteins across 5,262 connected components.
Component size is highly skewed (median 4; 95th percentile 20; max 107). Median components span two species, and only 18.5% contain more than 5 species.
Anomalous similarity is globally sparse: the median fraction of high-z edges per component is 0 and only 3.9% of components show substantial anomaly concentration ($high_z_frac > 0.3$). This sparsity is also evident in the global edge distribution (fig. 4), where most similarities cluster near the baseline while high-z edges occupy a small right tail.
Across 37,421 proteins in the pruned graph, only 5.9% satisfy the gating criteria and receive a non-zero score, indicating strong selectivity.

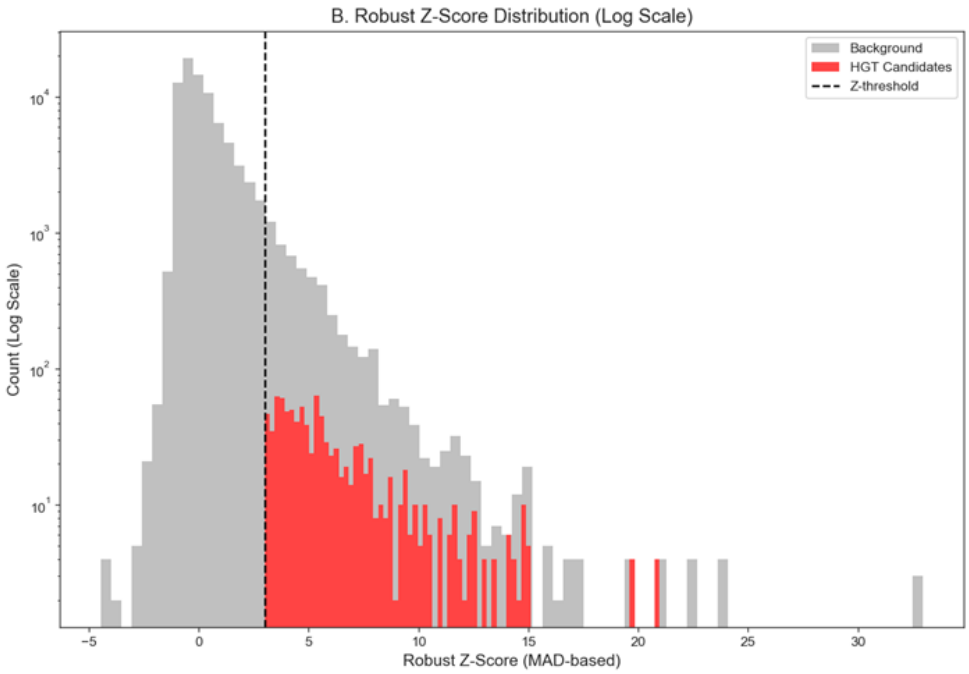

Figure 4: Distribution of MAD-based z-score for all protein-protein edges, plotted in logarithmic y-axis. The background distribution (grey) represents the global set of 911,306 edges, which clusters near the median, reflecting expected evolutionary divergence. The dashed black line is the statistical significance threshold (Z=3). Edges connected to the final HGT candidates (red) are localized to the right tail of the distribution.

4.2 Findings

We present below the top HGT candidates identified by each method, along with biological context supporting each finding.

4.2.1 Horizontal Transfer of EptC (Phosphoethanolamine Transferase)

Using method 1, we identified EptC as a candidate HGT event. EptC adds phosphoethanolamine (PEA) to the bacterial outer membrane, reducing its negative charge and thereby decreasing the efficacy of host antimicrobial peptides, providing a selective advantage. EptC is a known HGT-associated gene.[18]

In fig. 5 below we show a similarity graph in which a red thick line is connecting *Shigella* to two *Escherichia Coli* species. The low similarity of *Pseudomonas Aeruginosa* (cyan node) to the others suggests this is an HGT event rather than broad conservation. Notably, no suspicious edge appears between the two *E. coli* strains, thanks to taxonomic distance filtering.

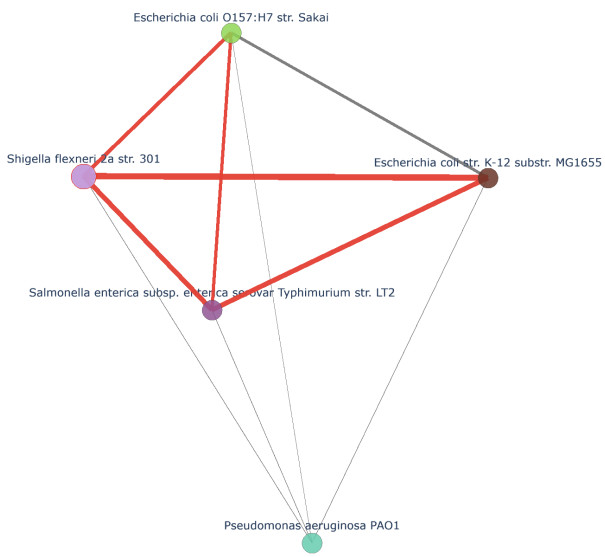

Figure 5: Similarity graph for the EptC gene. Each node denotes a protein and the text above denotes the origin species. The thicker the edge, the higher the similarity score between the proteins, A red edge hints at a potential HGT event.

Then, we turned to construct a gene phylogenetic tree to see which species are closest. As mentioned in step 4, trees were constructed using the NJ algorithm.[12] As expected from HGT, we got close relations between one strain of *E. coli*, *Shigella*, and *Salmonella*, differing from the species tree, which would join the two *E. coli* strains to each other.

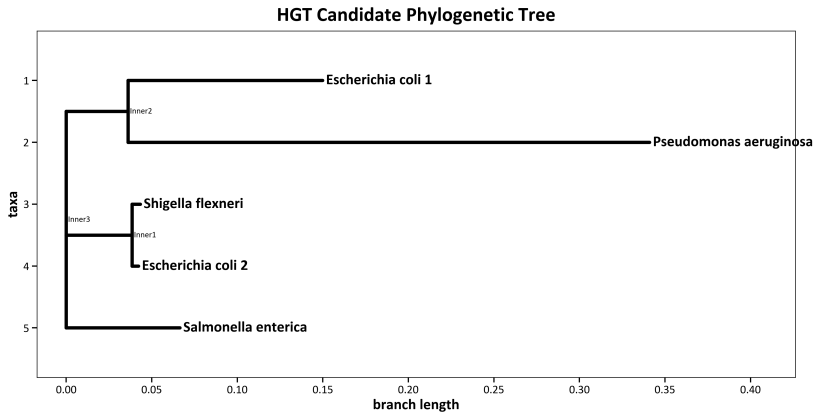

Figure 6: Constructed phylogenetic tree for the EptC coding gene. Note the abnormal distance between *E. coli* strains, suggesting HGT event.

4.2.2 Horizontal Transfer of ymdF

Using method 1, we identified ymdF as an additional candidate HGT event. At the time of writing this report, the exact function of the protein itself remains a mystery to the scientific community, although it is annotated as a stress-induced or stress-related protein.[19,20] Proteins involved in stress response can confer a selective advantage under adverse conditions, making such genes plausible candidates for horizontal acquisition. While there is no direct HGT evidence of ymdF itself, it has been proposed that the yciFE operon, which includes paralogs of ymdF, comprises horizontally transferred genes in both *E. coli* and *Salmonella*,[21] suggesting an HGT history of related genes within this operon. HGT is further supported again by our phylogenetic analysis. In fig. 7, the thick red edge connecting the corresponding *E. coli* and *Salmonella* nodes (purple and brown) clearly recovers the close relationship reported for the yciFE operon.[21] One node (gray) lacks species annotation in our dataset.

In fig. 8 we see the gene tree of ymdF. The tree suggests a close similarity between an *E.*

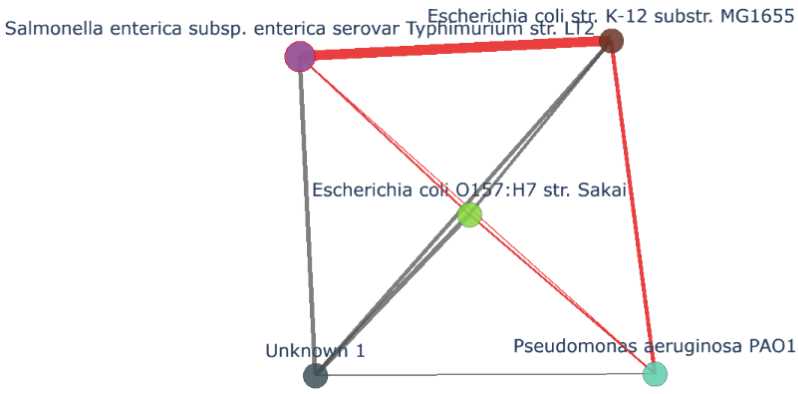

Figure 7: Similarity graph for the ymdF protein. The thicker the edge, the higher the similarity score between the proteins. A red edge hints at a potential HGT event.

coli strain and *Salmonella*, instead of the expected similarity between the two *E. coli* strains, supporting the clain of an HGT event.

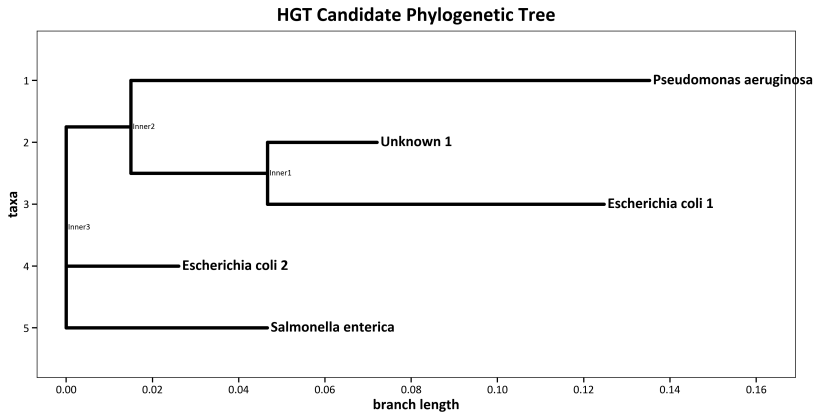

Figure 8: Constructed phylogenetic tree for the ymdF coding gene. Note the abnormal distance between *E. coli* strains, suggesting HGT event.

4.2.3 Horizontal Transfer of Glucose-1-phosphate thymidylyltransferase

Using method 2, we examined the Jaccard-pruned graph, focusing on connected components enriched for high surprise edges (z-score ≥ 3) and broad taxonomic mixing.

In connected component A (fig. 9), both clusters demonstrate a horizontal transfer of Glucose-1-phosphate thymidylyltransferase, a key building block for bacterial surface polysaccharides, which helps bacteria like *Capnocytophaga* (pathogenic oral commensal in dogs, purple node) resist the killing effect of human serum.[22,23]

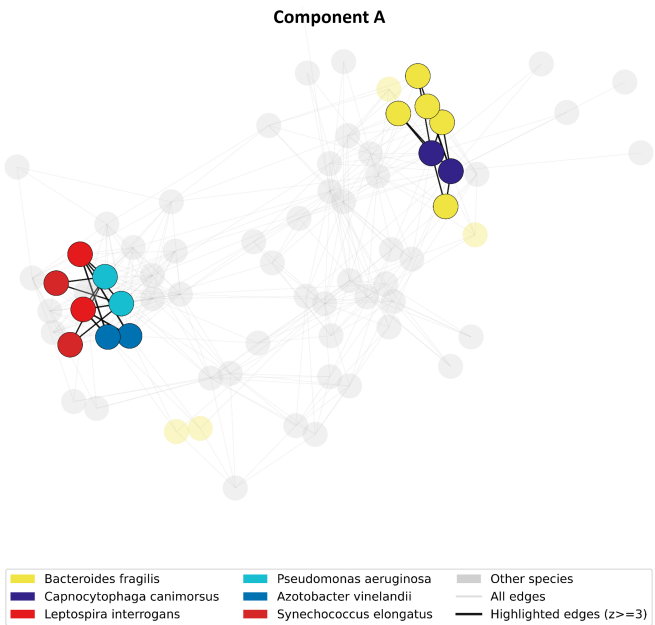

Figure 9: Connected component A of the Jaccard-pruned graph. Each node represents a protein sequence and is colored by species. Weighted edges connect protein with Jaccard similarity above the pruning threshold. Edge width reflects similarity strength. Bold edges denote statistically anomalous similarities (z-score ≥ 3).
Stats: #nodes=71, #edges=332.

4.2.4 Horizontal Transfer of 30S ribosomal protein S10 (rspJ)

In connected component B of the graph produced by method 2 (fig. 10), all clusters correspond to a horizontal transfer of 30S ribosomal protein S10 (rspJ gene). Mutations in this gene are associated with tigecycline resistance.[24] Studies specifically show the transfer of this gene involving tuberculosis and other Mycobacterium species.[25]

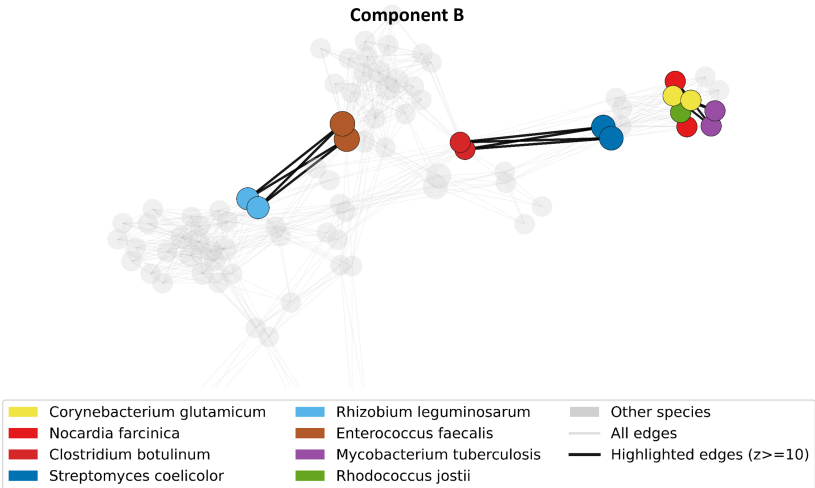

Figure 10: Connected component B of the Jaccard-pruned graph. Each node represents a protein sequence and is colored by species. Weighted edges connect protein with Jaccard similarity above the pruning threshold. Edge width reflects similarity strength. Bold edges denote statistically anomalous similarities (z-score ≥ 10).
Stats: #nodes=95, #edges=589.

4.2.5 Horizontal Transfer of Ribosomal Proteins S21 & S12 (rpsL)

Connected component C of the graph produced by method 2 (fig. 11) forms a densely connected subgraph comprising 100 proteins across 48 species. High scoring proteins appear throughout the network and interact across species, suggesting a shared module rather than isolated events. This pattern fits repeated horizontal gene transfer or shared mobile gene pools creating multi-species similarity.[26]

In cluster C_1 all nodes correspond to ribosomal protein S21, which is linked to late stage phage replication. Once the gene is acquired via HGT, it becomes a functional part of the phage's genetic toolkit for hijacking the host's ribosomes to favor phage protein synthesis during late infection.[27] This finding is consistent with results from other HGT detection tools.[28] Clusters C_2, C_3 correspond to ribosomal protein S12 (rpsL gene), whose mutations confer streptomycin resistance,[29] supporting our detection of an HGT event.

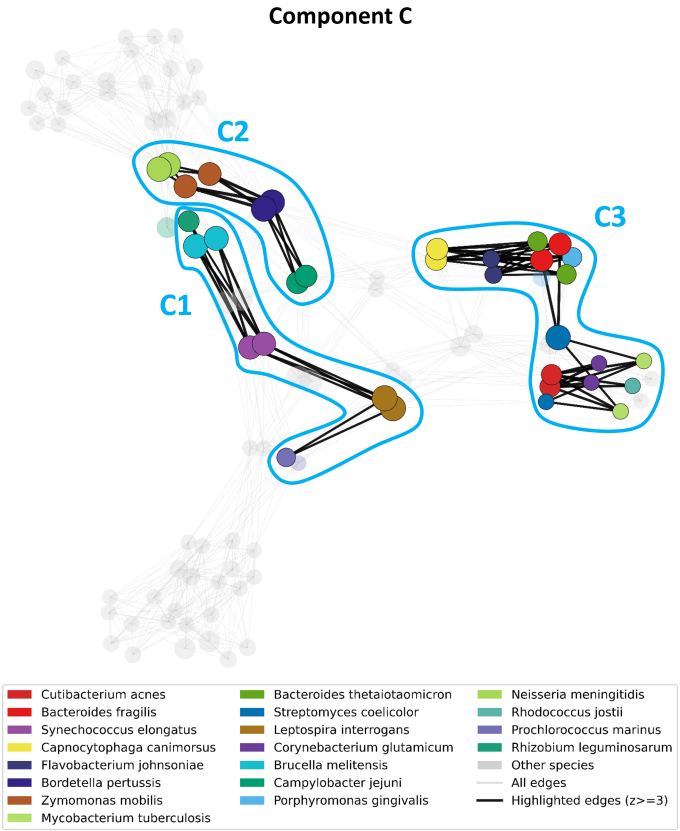

Figure 11: Connected component C of the Jaccard-pruned graph. Each node represents a protein sequence and is colored by species. Weighted edges connect protein with Jaccard similarity above the pruning threshold. Edge width reflects similarity strength. Bold edges denote statistically anomalous similarities (z-score ≥ 3).
Stats: #nodes=97, #edges=632.

5 Discussion

5.1 Summary

We asked whether graph-based analysis of cross-species protein similarity can reveal candidate HGT events. Our goal was not to replace classical phylogenetic approaches, but to develop a lightweight, interpretable framework using sparse similarity graphs, taxonomic context, and anomaly scoring.

Across both methodologies, we identified candidates whose similarity patterns deviate from species-level clustering expectations: EptC and ymdF using method 1, and Glucose-1-phosphate thymidylyltransferase, 30S ribosomal protein S10, and ribosomal proteins S21/S12 using method 2, supporting the utility of graph structure for exposing HGT candidates.

5.2 Analysis

Graph analysis provides meaningful information regarding HGT events: unusual edges become more informative when viewed in their broader network context.

In method 1, clustering proteins into homologous groups made the problem computationally manageable. Within these groups, suspicious edges connected proteins that were both highly

similar and taxonomically distant. Node-level aggregation then prioritized proteins repeatedly involved in anomalous relationships. Phylogenetic follow-up confirmed that inferred gene relationships did not match expected species relationships.

Method 2 shows that even without explicit alignments, large-scale k-mer-based graph screening can recover biologically interesting signals. The pruning, normalization and component-aware scoring narrowed a large candidate space to a small set of high-interest proteins, and high-scoring components displayed strong taxonomic mixing.

At the same time, our results should be interpreted carefully. They show that graph-based analysis is effective for prioritizing candidate HGT event, but not for proving an HGT event on its own. Similarity across taxa can also arise from conserved function, paralogy or annotation artifacts.

5.3 Evaluation

A strength of the project is that both methodologies addressed complementary formulations and both identified plausible HGT-related candidates, suggesting the graph-based perspective is robust.

The project has several limitations. First, similarity is not equivalent to horizontal transfer: our pipeline identifies candidates, not confirmed events. Second, taxonomic distance is coarse and does not fully model evolutionary history. Third, the two methods used different datasets, reducing direct comparability. Method 1 used a small set of proteomes, limiting taxonomic breadth; Method 2 used broader coverage but selecting at most two assemblies per species introduced some redundancy. Fourth, scoring thresholds were partly heuristic and different choices could change the ranked candidates.

A practical challenge was balancing biological realism with computational feasibility, motivating the divide-and-conquer designs. Another was distinguishing conserved proteins from truly anomalous cross-species similarity, addressed by combining similarity with taxonomic context and graph-level aggregation.

5.4 Future Work

The most valuable next step is validation against curated HGT benchmarks or established tools to enable quantitative precision/recall evaluation.

For scaling, sketching approaches such as MinHash[30] could reduce the cost of large-scale similarity screening in method 2.

The two methodologies could be unified into a single hierarchical pipeline: method 2's alignment-free construction as a broad screening layer, followed by method 1's alignment-based pipeline for finer analysis and phylogenetic inspection.

Finally, the framework could be extended beyond bacterial datasets to broader ecological or taxonomic boundaries.

6 Work Distribution Table

Name	Responsibility
Or Forshmit	Design and implementation of method 1. Visualizations for method 1.
Noam Kimhi	Pruning design of method 2. Cleaning visualizations. Coordination between the sub-teams. Presentation making.
Roee Tekoah	Implementation of method 2. Creation of visualizations for method 2.
Dor Stein	Implementation of method 1. Literature and biological validation of method 1's results.
Noam Korkos	Implementation of the pruning procedure for method 2. Validation of method 2's results.

Table 1: Team responsibilities

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